

UST-ILIMSK, Russian Federation

WMO: 301170

Lat: 58.030N

Lon: 102.720E

Elev: 1316

StdP: 14.01

Time Zone: 8.00 (E08)

Period: 90-01

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-36.6	-31.1	-42.3	0.5	-36.0	-35.5	0.8	-30.2	22.5	14.6	20.3	8.6	4.6	210	0.684

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.0	84.0	63.2	80.6	62.1	77.4	61.0	66.4	76.7	64.7	75.5	63.1	74.1	6.9	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.4	91.7	69.7	61.0	84.2	68.2	58.9	78.0	66.9	32.0	76.3	30.5	75.7	29.3	74.2	72.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.1	17.6	15.8	-36.6	89.4	5.4	1.7	-40.5	90.6	-43.6	91.6	-46.6	92.6	-50.5	93.8		
			-37.3	69.4	5.2	2.4	-41.0	71.1	-44.1	72.5	-47.0	73.8	-50.8	75.5		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	30.0	-5.5	3.5	16.1	31.0	46.3	58.2	66.8	60.8	43.8	30.3	8.2	-1.4
	DBStd	26.82	15.12	11.90	12.03	8.34	10.24	8.54	6.67	6.80	7.68	10.19	15.68	13.42
	HDD50	8544	1720	1302	1052	571	201	20	0	6	213	611	1255	1593
	HDD65	12967	2185	1722	1517	1021	585	239	62	163	635	1075	1705	2058
	CDD50	1244	0	0	0	1	86	266	521	341	28	1	0	0
	CDD65	192	0	0	0	0	6	35	118	33	0	0	0	0
	CDH74	1509	0	0	0	0	68	286	958	197	0	0	0	0
	CDH80	344	0	0	0	0	13	54	258	19	0	0	0	0
Wind	WSAvg	8.0	8.2	8.0	8.4	8.8	9.0	7.3	6.1	6.4	7.0	9.4	8.8	8.7
Precipitation	PrecAvg	14.3	0.7	0.5	0.5	0.8	1.3	1.8	2.3	2.1	1.5	1.1	1.0	0.9
	PrecMax	18.9	1.5	0.9	1.1	2.0	2.6	3.8	5.1	4.4	3.9	2.2	1.9	2.0
	PrecMin	11.5	0.3	0.1	0.0	0.1	0.0	0.4	0.4	0.2	0.2	0.2	0.4	0.4
	PrecStd	1.7	0.3	0.2	0.3	0.4	0.6	0.8	1.2	1.1	0.9	0.5	0.4	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	27.8	29.8	44.3	59.5	81.7	85.6	90.4	82.5	68.6	56.7	39.6	28.9
		MCWB	27.0	27.0	35.2	43.2	55.9	61.0	65.2	62.2	55.1	45.6	37.0	27.2
	2%	DB	23.8	25.4	38.5	53.5	75.0	81.1	85.7	78.9	62.2	50.6	34.9	23.8
		MCWB	22.9	23.2	31.2	39.8	53.4	60.1	65.2	62.2	51.8	41.9	33.1	22.9
	5%	DB	18.1	21.7	35.2	47.9	68.9	77.4	82.7	76.0	58.5	46.8	32.3	19.9
		MCWB	17.1	20.1	29.4	37.8	52.1	59.0	63.9	60.8	49.3	41.0	31.1	19.0
	10%	DB	13.5	18.4	32.0	43.2	63.7	73.5	79.5	73.1	54.8	42.8	29.1	16.0
		MCWB	12.7	17.0	27.3	34.8	49.1	57.7	62.9	59.9	47.8	37.6	27.5	15.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.9	27.0	35.4	44.8	58.5	64.6	69.9	66.4	55.8	46.3	37.4	27.4
		MCDB	27.5	29.0	42.5	56.9	76.5	77.9	80.6	71.9	64.6	52.5	40.1	28.6
	2%	WB	23.0	23.1	32.2	41.0	55.7	62.6	67.7	64.1	53.7	43.8	33.3	23.0
		MCDB	23.9	24.8	37.3	52.4	71.6	75.9	78.6	73.3	60.6	49.4	34.4	23.7
	5%	WB	17.1	20.1	30.0	38.1	53.4	60.8	66.2	62.7	51.2	41.2	31.1	19.1
		MCDB	18.0	21.6	34.7	46.5	66.1	73.2	76.7	73.5	56.3	46.1	32.5	19.8
	10%	WB	12.8	17.0	27.4	35.5	50.7	58.9	64.7	61.2	48.9	38.0	27.6	15.1
		MCDB	13.6	18.4	32.0	42.4	62.0	70.6	75.8	71.2	53.7	41.9	28.9	15.9
Mean Daily Temperature Range	5% DB	MDBR	10.0	12.8	15.4	13.8	16.7	17.0	17.0	15.6	12.7	11.1	11.2	11.8
		MCDBR	11.7	13.8	16.2	17.6	24.0	21.4	21.4	20.0	17.4	15.2	10.4	13.1
		MCWBR	11.3	12.7	11.4	11.1	12.6	9.2	8.0	8.0	8.8	9.9	9.6	12.9
	5% WB	MCDBR	12.0	13.2	14.9	16.1	21.0	19.7	18.2	16.7	14.8	13.4	10.2	13.2
		MCWBR	11.5	12.4	11.2	10.9	11.9	8.6	7.7	7.8	8.7	10.0	9.5	13.1
Clear-Sky Solar Irradiance	taub	0.272	0.301	0.325	0.359	0.385	0.423	0.459	0.411	0.345	0.328	0.283	0.255	
	taud	1.982	2.007	2.044	2.146	2.272	2.236	2.138	2.249	2.416	2.262	2.042	1.945	
	Ebn at Noon	167	222	254	267	269	259	244	249	252	216	161	132	
	Edh at Noon	19	29	37	40	38	40	44	36	26	23	17	14	
All-Sky Solar Radiation	RadAvg	202	509	976	1411	1632	1833	1699	1349	829	400	198	116	
	RadStd	22	38	58	98	124	142	136	85	102	41	23	20	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page